

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-6:

Aim: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

PROCEDURE :

1.Escape Sequence Handling and Command Parsing :

```
siddinapujith07@DESKTOP-FDHCAIT:~$ mkdir skill6
siddinapujith07@DESKTOP-FDHCAIT:~$ cd skill6
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ nano escape.c
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ gcc escape.c -o escape
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ ./escape
Enter input with escape characters: hello\ world

Original Input: hello\ world

Processed Output:
hello world

Parser output validated successfully.
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ ./escape
Enter input with escape characters: hello\@world

Original Input: hello\@world

Processed Output:
hello@world

Parser output validated successfully.
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ ./escape
Enter input with escape characters: hello\ world\@test

Original Input: hello\ world\@test

Processed Output:
hello world@test

Parser output validated successfully.
```

2.Child Process Creation and Program Execution :

```
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ nano execute.c
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ gcc execute.c -o execute
siddinapujith07@DESKTOP-FDHCAIT:~/skill6$ ./execute
Parent Process
Child Process
Parent PID = 1830
Child PID = 1831
Waiting for child...
total 40
-rwxr-xr-x 1 siddinapujith07 siddinapujith07 16216 Aug 22 14:04 escape
-rw-r--r-- 1 siddinapujith07 siddinapujith07 621 Aug 22 14:04 escape.c
-rwxr-xr-x 1 siddinapujith07 siddinapujith07 16304 Aug 22 14:06 execute
-rw-r--r-- 1 siddinapujith07 siddinapujith07 670 Aug 22 14:05 execute.c
Child process completed.
```

Code:

```
GNU nano 8.7.1          escape.c *
#include <stdio.h>
#include <string.h>

int main()
{
    char input[200];

    printf("Enter input with escape characters: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    printf("\nOriginal Input: %s\n", input);

    printf("\nProcessed Output:\n");

    for (int i = 0; input[i] != '\0'; i++)
    {
        if (input[i] == '\\' && input[i + 1] != '\0')
        {
            i++;
            printf("%c", input[i]);
        }
        else
        {
            printf("%c", input[i]);
        }
    }

    printf("\n\nParser output validated successfully.\n");

    return 0;
}
```

```
GNU nano 8.7.1          execute.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;
    int status;
    pid = fork();
    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }
    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());

        execlp("ls", "ls", "-l", NULL);

        perror("Execution failed");
        exit(1);
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Waiting for child...\n");
        waitpid(pid, &status, 0);
        printf("Child process completed.\n");
    }
    return 0;
}
```

Observation :

1. Escape characters were identified and processed correctly.
2. Escaped spaces and special symbols were preserved as literal characters.
3. A child process was successfully created using fork().
4. The child process executed the ls program with arguments.
5. The parent process waited for the child using waitpid().
6. Invalid program execution was detected and handled using an error message.

Result :

The programs successfully demonstrated **escape-sequence processing and child-process execution**. Escaped characters were handled correctly, while the child process successfully launched external programs, passed arguments, reported execution errors, and synchronized with the parent process.